DNA: the raw material of genetics

24A

Genetics is the study of inheritance — how characteristics are passed from one generation to the next. Genetic material is stored in the nucleus of each cell. The major bodies involved in this process are called chromosomes. Each chromosome is made up of genes. Genes are the instructions that are responsible for the components and appearance of an organism. They are found in the nucleus of a cell.

> **A gene**: the part of a chromosome that determines the inheritance of a characteristic or of a group of characteristics.

Each chromosome is made of a single molecule of DNA. The DNA molecule is a chain of many genes connected together. DNA is a nucleic acid and is a very large molecule. If the DNA in a single cell was uncoiled and put in a straight line, it would be a metre long. The backbone of the DNA molecule is two chains wound around each other, forming a shape rather like a twisted ladder, called a double helix. There are a large number of smaller molecules (amino acids) attached. The order in which these side groups are attached determines the code of a particular nucleic acid. The code tells the cell how to reproduce itself. Cells divide in an organism for growth, repair and reproduction.

Watson, Crick and Wilkins shared the 1962 Nobel Prize for their contributions to the understanding of the genetic code. Watson and Crick came up with our current double helix (spiral) model of DNA. In humans each DNA molecule is thought to contain from a few to many thousand genes. Each gene can be thought of as a series of 'rungs' on the DNA 'ladder'.

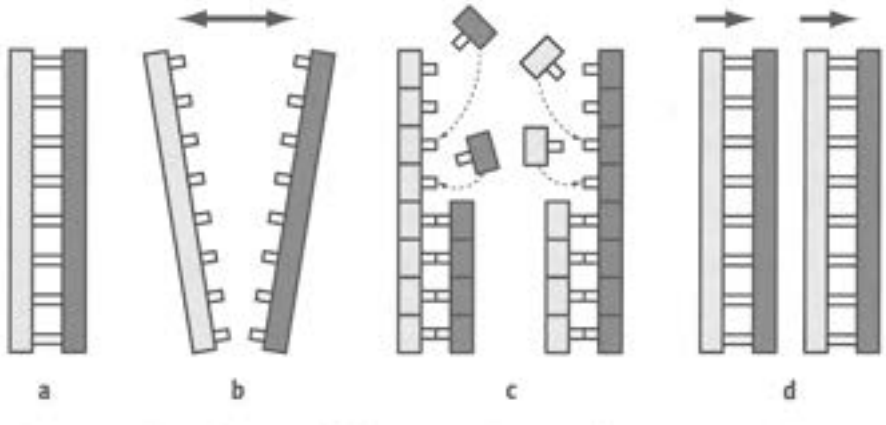
A simplified version of DNA replication is shown.

a The double strand of DNA.

b The double strand 'unzips' and two single strands are formed.

c Both strands (chromatids) are replicated (copied).

d The strands join to form two double strands identical to the original chromosome.



DNA is made up of many nucleotides. A nucleotide is a little unit made up of a sugar, at least one phosphate and a nitrogen base. The nitrogen base contains a ring (or rings) containing nitrogen. Each half of the strand contributed by a parent contains many alleles. Alleles are particular (non-identical) forms of a gene, coding for a particular trait. The alleles can be compared to individual teeth on a zipper. When each tooth of the zipper pairs with its correct mate, the zipper is complete. Similarly, when each allele pairs up with its correct mate, a gene pair is formed.

The exact location of an allele on a strand of DNA is called its locus.

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DNA: the raw material of genetics

24B

It is important that the DNA is copied exactly. Otherwise information to make new cells will be incomplete or faulty. Any mistakes will be passed on to the next generation of cells. These cells could have problems that could cause the organism to be defective or less successful in its environment.

Genes are not always copied exactly. Mutations can occur — these alter genes.

> **A mutation:** a change in a gene (or genes). The resulting mutated gene may result in a different phenotype (physical characteristic).

Mutations can involve the sequence of genes being rearranged or changed. If these changes occur in gametes (sex cells), they can be inherited. If they are recessive (genes which do not automatically show up in physical characteristics), they may go undetected. These mutations may result in diseases, such as a blood disease called sickle-cell anaemia. Mutations can be favourable or unfavourable to a species. They lead to changes of the characteristics of individuals in the population of a group of organisms. If favourable, these changes may make those individuals more likely to survive and breed. If unfavourable, these changes may make those individuals less likely to survive and breed, so they will die out.

Mutations very rarely happen spontaneously. They are usually caused by mutagens such as:

- chemicals (acridine dyes, fungi that cause food to rot, nitrous acid, tobacco tars)
- heat (increasing temperature can cause a rise in mutation rate)
- radiation (cosmic rays, gamma rays, ultraviolet [UV] light and X-rays)

1 How many strands make up a DNA molecule? _____

2 What is the shape of a DNA molecule? _____

3 'A gene is a short length of DNA'. True or false? _____

4 Why do cells divide? _____

5 Why is it important that DNA is copied exactly during replication? _____

_____

6 Who is credited with the current model of DNA? _____

7 What does the code of a nucleic acid tell a cell to do? _____

8 What happens in mutation? _____

9 Name two causes of mutation. _____

10 Are mutations always a disadvantage? _____

11 What is a locus? _____

12 What is a nucleotide? _____

13 What proportion of genetic information does a child inherit from each parent? _____

14 What is a 'chromatid'? _____

15 Give a definition of 'genetics'. _____

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